

Economic Scheduling of the UAN Cogeneration Plant: A Preliminary Optimization Study

ALI M. EL-NASHAR and M. SARFRAZ KHAN

*Cogeneration Analysis Section, Water and Electricity Department, P.O. Box 219,
Abu Dhabi (UAE)*

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SUMMARY

In this paper, an attempt is made to solve the unit scheduling problem as applied to the cogeneration complex at Umm Al nar station. This complex supplies most of the electrical power and desalinated water needs of the city of Abu Dhabi, UAE. The objective of the study is to find the optimal load distribution among the different power generating units which would yield a minimum fuel energy consumption for a given station power and water demands. The paper shows how this problem is solved under purely static operating conditions assuming a constant water supply of desalinated water by MSF distillers. To simplify the problem, the UAN station is first divided into five production block with the operation of each block optimized individually. Subsequently, the optimization of the five blocks is carried out to obtain the load which should be carried by each block for a minimum energy consumption.

Keywords: unit scheduling, cogeneration, optimal operation, economic dispatch

Abbreviations: EHR — exhaust heat recovery, D — MSF distiller, GT — gas turbine, HP — high pressure, MP — medium pressure, MSF — multistage flash, LP — low pressure, UAN — UmmAl Nar, UANE — Umm Al Nar East, UANW — Umm Al Nar West

LIST OF SYMBOLS

- n — operating status of a unit having a value 0 or 1
 P — power generated, MW
 Q — fuel energy consumption, GJ/h
 S — steam flow rate, t/h
 W — distillate flow rate, t/h

Subscripts

- ab — pertaining to auxiliary boiler
 b — referring to main boiler
 bk — related to a block
 brt — brine recycle turbine
 d — distiller
 dm — pertaining to MP steam for a distiller
 ej — pertaining to steam ejector
 ext — pertaining to steam extraction
 gt — related to gas turbine
 red — pertaining to reducing station
 rh — pertaining to steam turbine reheat steam
 rb — related to EHR boiler
 st — pertaining to steam turbine
 sf — related to supplementary firing
 t — pertaining to the whole UAN station

INTRODUCTION

Cogeneration for power and fresh water production has been practiced by utilities in the Middle East for many years as a reliable, efficient, and economic means of generating power and desalting sea water. It offers savings in fuel consumption due to the reduced amount of heat rejected to the environment resulting from the dual production of electricity and process heat in one plant as compared with two separate plants producing the same amount of power and water.

Cogeneration plants of this type are normally connected to the national grid and usually operate in an electric-load-following mode. This means that the plant is operated in such a way that the power generated is allowed to vary so as to keep the variation of the grid frequency within a tightly controlled range. The desalination units associated with the plant, on the other hand, generally operate at steady load (normally full load), with only occasional changes in their production levels. This mode of operation is dictated by the fact that for most cogeneration plants a certain amount of water storage capacity is installed with the plant.

The energy demand in any cogeneration system shows large daily and seasonal variations. This poses the problem of how to determine which

units should be running and which ones should be shutdown. This is a unit scheduling problem, which, in fact, combines two mutually connected problems. The first one is usually called the unit commitment problem, and this consists of the optimum selection of the units that must be put into operation and the ones that should be shutdown. The second is usually called the economic dispatch problem, and this deals with the allocation of the load among the different units in operation such that to minimize a certain criteria, such as the operating cost of the plant, subject to certain constraints relating to the demand and spinning reserve. Experience shows that proper scheduling of generating units can result in fuel cost savings which can be substantial.

While methods of solving the economic dispatch problem has been well known for a long time [1,2,3], the unit commitment problem has proved to be more challenging, particularly for a general cogeneration system with demand variation in both power and water. Fortunately in a typical cogeneration system, the demand for one product (water) can be safely regarded as constant since there usually exists enough water storage capacity to act as a buffer between a variable water demand and an essentially constant water supply by the MSF distillers.

In this paper, an attempt is made to solve the unit scheduling problem as applied to the cogeneration complex at Umm Al Nar station. This complex supplies most of the electrical power and fresh water needs of Abu Dhabi city. As this is a preliminary study, the problem is solved in a somewhat idealistic setting in which all dynamic constraints related to the down-time and up-time of all the operating units, as well as spinning reserve and network security considerations are not included. Furthermore, the operating desalination units are assumed to be running at full load producing a constant amount of fresh water. Therefore, we essentially consider a system which has to meet a variable demand for electricity and, at the same time, produce a given amount of fresh water, and the problem is to determine appropriate outputs of the power generating units so as to minimize the cost of satisfying these demands.

DESCRIPTION OF UAN STATION

The Umm Al Nar (UAN) station is the main cogeneration complex that provides most of the power and desalinated water needs of the Abu Dhabi city. The station has an installed capacity of 1100 MW of electrical power and 14,570 t/h (with polyphosphate antiscalant) of desalinated water. Four gas turbines, and 10 steam turbines having widely different designs and rated capacities are installed for power generation. Desalination of sea water is accomplished by 16 multistage flash (MSF) units of varying performance ratio and rated distillate production capacities. The station is

physically divided into two main plants based on their location with respect to north direction, namely UAN East plant and UAN West plant. Each plant is briefly described in the following sections.

The UAN East plant

The UAN East plant has a rated power production capacity of 270 MW and a rated distilled water production of 6,270 t/h. The plant consists of four gas turbines, four exhaust heat recovery boilers, six auxiliary boilers and six MSF distillers. The plant can be conveniently divided into two sections with each section forming an independent plant by itself, but can also be joined together to form a single plant. The two sections are given the names UAN East and UAN East (extension) since the later plant is simply an expansion of the former one.

The UAN East section comprises two gas turbines, two EHR boilers, four auxiliary boilers and three distillers. The gas turbines are manufactured by Brown Boverie Company and each has a peak output power of 65 MW and a rated power of 62.5 MW at 15°C ambient temperature (about 60 MW at the average ambient temperature in Abu Dhabi).

The two EHR boilers has a steam raising capacity of 186 t/h at a pressure of 16 bar and a temperature of 210°C when receiving 1166 t/h of turbine exhaust gases at a temperature of 514°C. These boilers do not have the facility of supplementary firing in the exhaust duct. Each of the four auxiliary boilers has a rated capacity of 136 t/h of steam at 16 bar and 210°C and can use either natural gas or reduced crude oil as fuel. Both the EHR and auxiliary boilers are manufactured by Borsig Company.

The three MSF distillers are manufactured by Sidem Company and each has a rated capacity of 950 t/h of distillate and a performance ratio of 6.1 when using polyphosphate as the antiscalant. Each distiller has a small single-stage steam turbine to drive its brine recirculating pump.

The UAN East (ext.) section, on the other hand, consists of two gas turbines, two EHR boilers, two auxiliary boilers and three MSF distillers. The gas turbines are Type TG50 manufactured by Fiat Company and have a rated power output of 75 MW at an ambient temperature of 30°C.

The two EHR boilers each has a steam raising capacity of 210 t/h at 16 bar and 210°C and has the provision of supplementary firing which can provide up to 107 t/h of additional steam for the distillers when the turbines are running at part load. The two auxiliary boilers has a rated capacity of 160 t/h of steam at the same conditions as the EHR boilers. All the EHR and auxiliary boilers of UAN East (ext.) are also of Borsig manufacture. Either natural gas or reduced crude oil can be used as fuel for both auxiliary boilers and supplementary firing.

The three MSF distillers at UAN East (ext.) are manufactured by Italimpianti Company and each has a rated capacity of 1140 t/h of distillate and a performance ratio of 7.1 when using polyphosphate as the antiscalant. The brine recirculation pumps for each unit has an electric drive.

It should be mentioned here that all the steam generated by the boilers of both sections of UAN East plant are fed to two steam headers which have isolating valves that enable the two sections either to operate as a single unit, or as two separate and independent units. The normal mode of operation is to close these valves thus isolating the two units and operating each one independently of the other.

The UAN West plant

The ten steam turbines currently in operating at UAN West plant constitute a wide variety of power rating and designs. The power rating varies from 60 MW to 160 MW and their design varies from a purely condensing system to a reheat/extraction design. Six of the turbines (named UAN West 1-6) are controlled extraction/condensing machines having a rated capacity of 60 MW and are capable of accommodating large electrical load changes while maintaining constant extraction pressure. These six units operate on steam having a throttle condition of 60 bar and 485°C and are manufactured by Skoda Company. Each of the six units has an associated distiller which is supplied by LP steam from the turbine extraction port for the brine heater, and by HP steam directly from the boiler to drive its brine recirculating pump turbine.

Each of the associated distillers, which are manufactured by IHI Company, has a rated capacity of 750 t/h of distillate at a performance ratio of 6.0 using polyphosphate as the antiscalant chemical. The brine recirculating pumps of these distillers are driven by small single-stage steam turbines which consume about 33 t/h of HP steam to drive each pump.

Two of the turbines are reheat/extraction machines (named UANW 7 and 8 and manufactured by Skoda) having a rated capacity of 160 MW and using steam throttle conditions of 130 bar and 510°C. Each machine is capable of supplying enough extraction steam (286 t/h) for the operation of two distillers associated with it. Each of these distillers has a rated capacity of 950 t/h of distilled water at a performance ratio of 7.0 when using polyphosphate as antiscalant. These distillers are supplied by Sidem.

The last two steam turbine units at UAN plant (referred to as UANW 9 and 10 and manufactured by Franco Tosi Company) are purely condensing units having a rated capacity of 75 MW each with their throttle steam conditions being 87 bar and 510 °C.

OPTIMIZATION OF THE OPERATING CONDITIONS OF THE BLOCKS

Optimization procedure

The aim of an operation optimization scheme for a cogeneration system for power and fresh water production is to allocate the loads on the turbines and distillers in such a way that meets the demands for power and water, and, at the same time, achieves this with the minimum possible fuel consumption and an adequate spinning reserve to meet unexpected eventualities. This problem is dual in nature. First, unit commitment decisions has to be made in order to establish the units which should be on and those which should be off, for each feasible load combination. Then, the loads are allocated in an economic manner between the operating units in order to achieve the minimum fuel consumption. This problem has been discussed extensively in the literature, see for example references 2,3,4 and 5. For a given electrical and water demand, the total fuel energy consumption Q_t required to meet these demands may be expressed as

$$Q_t = \sum_{i=1}^{N_1} Q_{bi} + \sum_{i=1}^{N_2} Q_{abi} + \sum_{i=1}^{N_3} Q_{sfi} \quad (1)$$

where N_1, N_2 and N_3 are, respectively, the number of main boilers, auxiliary boilers and supplementary firings in operation. This total energy consumption will also depend on the loads carried by the individual power generating units as well as the MSF distillers. Thus we may write

$$Q_t = Q_t(P_{gt1}, P_{gt2}, \dots, P_{gt4}, P_{st1}, P_{st2}, \dots, P_{st10}, W_{d1}, W_{d2}, \dots, W_{d16}) \quad (2)$$

The minimization of fuel energy consumption required to produce certain demands of electrical power P_t and fresh water W_t may formally be represented by:

$$\min Q_t(P_{gt1}, P_{gt2}, \dots, P_{gt4}, P_{st1}, P_{st2}, \dots, P_{st10}, W_{d1}, W_{d2}, \dots, W_{d16}) \quad (3)$$

subject to the constraints

$$\begin{aligned} \sum_{i=1}^4 P_{gti} + \sum_{i=1}^{10} P_{sti} &= P_t \\ \sum_{i=1} W_{di} &= W_t \\ P_{gti_{\min}} &\leq P_{gti} \leq P_{gti_{\max}} \\ P_{sti_{\min}} &\leq P_{sti} \leq P_{sti_{\max}} \\ W_{di_{\min}} &\leq W_{di} \leq W_{di_{\max}} \end{aligned} \quad (4)$$

The first two constraints are equality constraints which set the summation of the power generated by the individual turbines, and the water produced by the individual distillers to the corresponding demands for power and water for the whole of UAN station. The remaining constraints are inequality specifying the limits on the load of all the power generating units and distillers in operation.

In order to simplify the problem, the system is divided into five subsystems or production blocks [6,7]. Each block is optimized individually for given demands of power and water to yield unit schedules, load distribution, and minimum fuel energy requirements corresponding to these demands. Subsequently, the results from the different production blocks are combined together in a higher-level optimization aimed at finding the optimum load allocation between the different blocks, see Fig. 1. Therefore, system op-

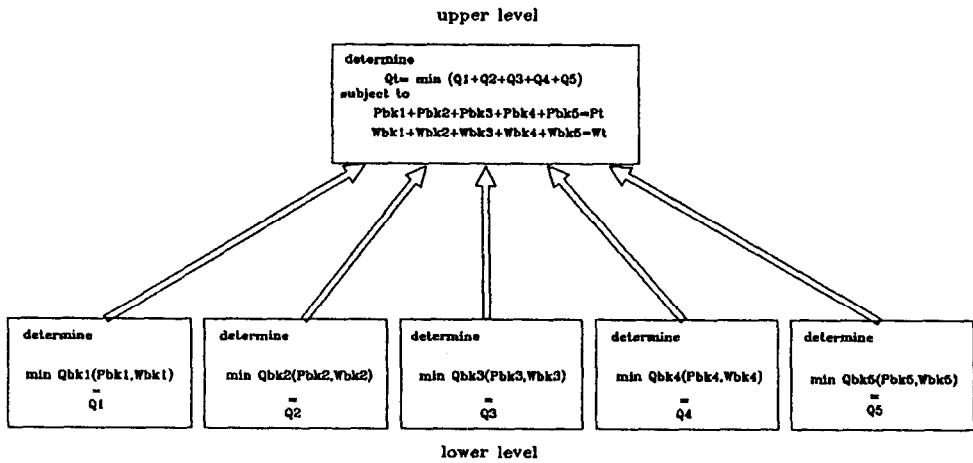


Figure 1. Optimization procedure used in the study.

timization is carried out on two levels, the lower level corresponds to the block optimization problem with the higher-level optimization joining the results of all the block optimization together. The lower-level objective functions and associated constraints can be written as

$$Q_{bki}(P_{bki}, W_{bki}) = \min Q_{bki}(P_1, P_2, \dots, P_i, W_{d1}, W_{d2}, \dots, W_{di}) \quad (5)$$

subject to

$$P_1 + P_2 + \dots + P_i + \dots = P_{bki}$$

$$W_{d1} + W_{d2} + \dots + W_{di} + \dots = W_{bki}$$

$$\begin{aligned} P_{i_{\min}} &\leq P_i \leq P_{i_{\max}} \\ W_{dj_{\min}} &\leq W_{dj} \leq P_{dj_{\max}} \end{aligned} \quad (6)$$

where P_{bki} and W_{bki} are the power generated and water produced by production block i .

Table I shows the identity of each block and the gas turbines, steam turbines and distillers in use. It should be noted that blocks 1–4 are essentially cogeneration plants with associated MSF distillers, whereas Block 5, identified as UANW 9 and 10, is basically a single purpose, power-only plant.

UAN East (Block 1)

A schematic diagram of this block is shown in Fig. 2. The fuel energy consumed by Block 1 for a given load combination of power and water may be expressed as

$$Q_{bk1} = \sum_{i=1}^2 Q_{gti} + \sum_{i=1}^4 Q_{abi} \quad (7)$$

In order to incorporate the on-off operating status of any turbine, boiler or distiller in the objective function, each machine has associated with it a discrete variable n that indicates whether or not it is running. It can have the value 1 if the machine is on and the value 0 if it is off. This variable is multiplied by the constant coefficient in the input-output correlation for the machine.

TABLE I
Identity of the production blocks.

Block number	Block identity	Gas turbines in use	Steam turbines in use	Distillers in use
1	UANE	GT 1 and 2	none	D1–3
2	UANE (ext.)	GT 3 and 4	none	D4–6
3	UANW 1–4	none	ST 1–4	D7–12
4	UANW 5–8	none	ST 5–8	D13–16
5	UANW 9 & 10	none	ST 9 & 10	none

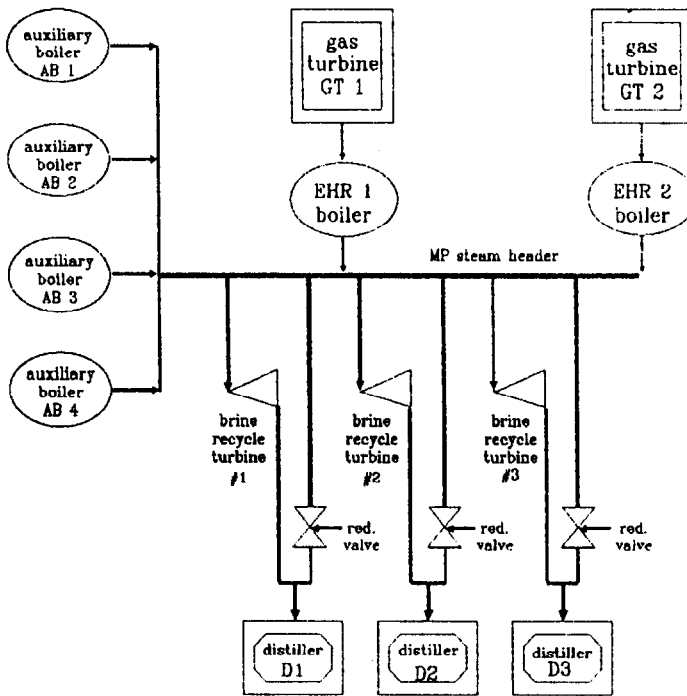


Figure 2. Simplified schematic of Block 1.

Expressing Q_{gt} as a polynomial function of a second degree in the power generated P_{gt} and Q_{ab} as a linear function of the steam flow rate we can write

$$Q_{bk1} = \sum_{i=1}^2 (a_{gti}n_{gti} + b_{gti}P_{gti} + c_{gti}P_{gti}^2) + \sum_{i=1}^4 (a_{abi}n_{abi} + b_{abi}S_{abi}) \quad (8)$$

The steam produced by the auxiliary boilers is related to the water produced by the distillers and the load on the gas turbines in operation. To obtain this relationship, a simple steam mass balance is made on a typical distiller. It should be noted here that each of the three distillers of UANE section has a recycle single-stage steam turbine driving the brine recycle pump associated with the MSF distiller. This turbine is driven by medium pressure steam (about 56 t/h) directly supplied from the MP steam header, see Fig. 2. Furthermore, a small amount of MP steam (about 5 t/h) is also needed to drive the steam ejectors. However, the major steam supply to the distiller is provided to the brine heater of the distiller (about 156 t/h at full-load) at a pressure of about 1.5 bar. Part of this amount is supplied

in the form of discharge steam from the brine recycle turbine, and the rest is provided from a reducing station. The amount of MP steam required by a distiller is given by

$$\begin{aligned}
 S_{dm} &= S_{red} + S_{brt} + S_{ej} \\
 &= \frac{(a_d n_d + b_d W_d) - S_{brt}}{(1 + \alpha)} + S_{brt} + S_{ej} \\
 &= a_{dm} n_d + b_{dm} W_d
 \end{aligned} \tag{9}$$

where S_{dm} is the amount of MP steam supplied to a distiller, S_{red} is the amount of MP steam supplied to the reducing station of a distiller, S_{ej} is the MP steam flow rate to an ejector and α is the ratio of desuperheating water flow rate to the MP steam flow rate to the ejector. The constants a_{dm} and b_{dm} assume the values: $a_{dm} = 29.3$, $b_{dm} = 0.134$. The amount of MP steam required by the three distillers has to be supplied jointly by the exhaust heat recovery boilers and the auxiliary boilers. Thus we can write the amount of steam required to be produced by the four auxiliary boilers as

$$\begin{aligned}
 \sum_{i=1}^4 S_{abi} &= \sum_{i=1}^3 S_{dmi} - \sum_{i=1}^2 S_{rbi} \\
 &= \sum_{i=1}^3 (a_{dmi} n_{di} + b_{dmi} W_{di}) - \sum_{i=1}^2 (a_{rbi} n_{rbi} + b_{rbi} P_{gti})
 \end{aligned} \tag{10}$$

The fuel energy supplied to the boiler can now be written as

$$Q_b = a_b n_b + b_b S_b \tag{11}$$

The fuel energy is zero when the boiler is off ($n_b = 0$) and has the value given by the correlation when it is on ($n_b = 1$). The objective function may now take the form

$$Q_{bk1} = \sum_{i=1}^2 (a_{gti} n_{gti} + b_{gti} P_{gti} + c_{gti} P_{gti}^2) + \sum_{i=1}^4 (a_{abi} n_{abi} + b_{abi} S_{abi}) \tag{12}$$

subject to

$$\begin{aligned}
 \sum_{i=1}^4 S_{abi} &= \sum_{i=1}^3 (a_{dmi} n_{dmi} + b_{dmi} W_{dt}) - \sum_{i=1}^2 (a_{rbi} n_{rbi} + b_{rbi} P_{gti}) \\
 P_{gti_{\min}} &\leq P_{gti} \leq P_{gti_{\max}} \\
 W_{di_{\min}} &\leq W_{dt} \leq W_{di_{\max}} \\
 \sum_{i=1}^2 P_{gti} &= P_{bk1}
 \end{aligned} \tag{13}$$

$$\sum_{i=1}^3 W_{dt} = W_{bki}$$

$$n_{gti} = 0 \text{ or } 1$$

$$n_{abi} = 0 \text{ or } 1$$

$$n_{rbi} = 0 \text{ or } 1$$

The optimization problem was solved by allowing each of the discrete variables to assume their extreme values of 0 and 1 then solve the resulting simplified problem using an appropriate computer algorithm that is commercially available. A number of these simplified problems were solved for all the possible combinations and the solution yielded the minimum fuel energy consumption was taken as the final solution to our optimization problem for Block 1.

UAN Est (ext.) (Block 2)

A simplified schematic of this block is shown in Fig. 3. The fuel energy consumed by Block 2 for a given load combination of power and water may be expressed as

$$Q_{bk2} = \sum_{i=1}^2 Q_{gti} + \sum_{i=1}^2 Q_{abi} + \sum_{i=1}^2 Q_{sfi} \quad (14)$$

Substituting polynomial expressions for Q_{gti} , Q_{abi} and Q_{sfi} into the above equation we get

$$\begin{aligned} Q_{bk2} = & \sum_{i=1}^2 (a_{gti}n_{gti} + b_{gti}p_{gti} + c_{gti}P_{gti}^2) + \sum_{i=1}^2 (a_{abi}n_{abi} + b_{abi}S_{abi}) \\ & + \sum_{i=1}^2 (a_{sfi}n_{sfi} + b_{sfi}S_{sfi}) \end{aligned} \quad (15)$$

If we notice that the three distillers of UANE (ext.), D4, D5 and D6, have electrically driven brine recirculation pumps, we can then write the amount of MP steam required by a single distiller as

$$\begin{aligned} S_{dm} &= S_{red} + S_{ej} \\ &= \frac{(a_d n_d + b_d W_d)}{(1 + \alpha)} + S_{ej} n_d \\ &= a_{dm} n_d + b_{dm} W_d \end{aligned} \quad (16)$$

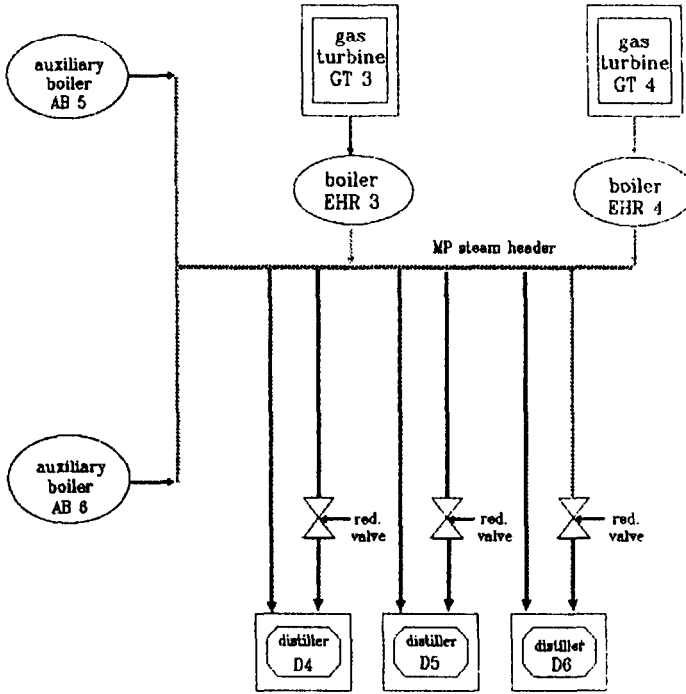


Figure 3. Simplified schematic of Block 2.

The amount of MP steam required by the three distillers must equal the amount of steam produced by the exhaust heat recovery boilers (from both exhaust gases and supplementary firing) and the auxiliary boilers. Therefore we can write this steam mass balance equation as

$$\begin{aligned}
 \sum_{i=1}^3 S_{dmi} &= \sum_{i=1}^2 S_{rbi} + \sum_{i=1}^2 S_{sfi} + \sum_{i=1}^2 S_{abi} \\
 &= \sum_{i=1}^2 (a_{rbi} + b_{rbi} P_{gti}) + \sum_{i=1}^2 S_{sfi} + \sum_{i=1}^2 S_{abi}
 \end{aligned} \quad (17)$$

The objective function for Block 2 may now be written as

$$\begin{aligned}
 Q_{bk2} &= \sum_{i=1}^2 (a_{gti} n_{gti} + b_{gti} P_{gti} + c_{gti} P_{gti}^2) + \sum_{i=1}^2 (a_{abi} n_{abi} + b_{abi} S_{abi}) \\
 &\quad + \sum_{i=1}^2 (a_{sfi} n_{sfi} + S_{sfi})
 \end{aligned} \quad (18)$$

subject to the following constraints

$$\begin{aligned}
 \sum_{i=1}^3 (a_{dmi}n_{di} + b_{dmi}W_{di}) &= \sum_{i=1}^2 (a_{rbi}n_{rbi} + b_{rbi}P_{gti}) + \sum_{i=1}^2 S_{sfi} + \sum_{i=1}^2 S_{abi} \\
 P_{gti_{\min}} &\leq P_{gti} \leq P_{gti_{\max}} \\
 W_{di_{\min}} &\leq W_{di} \leq W_{di_{\max}} \\
 S_{abi_{\min}} &\leq S_{abi} \leq S_{abi_{\max}} \\
 S_{sfi_{\min}} &\leq S_{sfi} \leq S_{sfi_{\max}} \\
 \sum_{i=1}^2 P_{gti} &= P_{bk2} \\
 \sum_{i=1}^3 W_{di} &= W_{bk2} \\
 n_{di} &= 0 \text{ or } 1 \\
 n_{rbi} &= 0 \text{ or } 1
 \end{aligned} \tag{19}$$

UAN West (Block 3)

A simplified schematic of this block is shown in Fig. 4. The fuel energy required by Block 3 to produce P_{bk3} MW of power and $W_{bk3}t/h$ of distillate is simply the fuel energy consumed by the four main boilers of UANW 1–4 thus

$$\begin{aligned}
 Q_{bk3} &= \sum_{i=1}^4 Q_{bi} \\
 &= \sum_{i=1}^4 (a_{bi}n_{bi} + b_{bi}S_{bi})
 \end{aligned} \tag{20}$$

The steam produced by these boilers should be sufficient to generate the power required by the steam turbines and also to provide the distillers in operation with their steam demand. Each distiller requires a quantity of high-pressure steam to drive the brine recycle turbine and the steam ejector, but the bulk of the steam requirement is at low pressure. The supply of high-pressure steam is taken directly from the boiler, whereas the low-pressure steam can be supplied from turbine extraction and/or from the reducing stations.

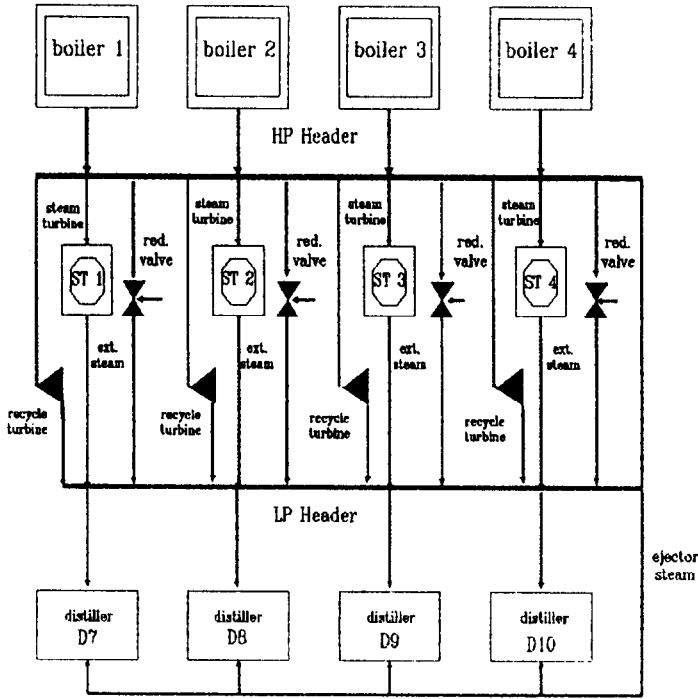


Figure 4. Simplified schematic of Block 3.

The amount of LP steam required by the four distillers is

$$\begin{aligned}
 \sum_{i=1}^4 S_{di} &= \sum_{i=1}^4 (a_{di}n_{di} + b_{di}W_{di}) \\
 &= \sum_{i=1}^4 S_{exti} + \sum_{i=1}^4 (1 + \alpha)S_{redi} + \sum_{i=1}^4 S_{brti}
 \end{aligned} \tag{21}$$

The amount of HP steam required by the four distiller is

$$\begin{aligned}
 \sum_{i=1}^4 S_{dhi} &= \sum_{i=1}^4 S_{redi} + \sum_{i=1}^4 S_{brti} + \sum_{i=1}^4 S_{eji} \\
 &= \frac{1}{(1 + \alpha)} \left\{ \sum_{i=1}^4 (a_{di}n_{di} + b_{di}W_{di}) - \sum_{i=1}^4 S_{exti} - \sum_{i=1}^4 S_{brti} \right\} \\
 &\quad + \sum_{i=1}^4 S_{bpi} + \sum_{i=1}^4 S_{eji}
 \end{aligned} \tag{22}$$

where α is the amount of desuperheating water expressed as a fraction of the HP steam flow rate entering a reducing station

The amount of HP steam required by the four steam turbines depends on both the electrical power to be generated as well as the amount of extraction steam supplied by the turbines. Thus we can write

$$\sum_{i=1}^4 S_{sti} = \sum_{i=1}^4 (a_{sti}n_{sti} + b_{sti}P_{sti} + c_{sti}P_{sti}^2 + d_{sti}n_{sti}S_{exti} + e_{sti}P_{sti}S_{exti}) \quad (23)$$

Therefore the total amount of HP steam produced by the four main boilers can be obtained by adding the HP steam for the distillers to that of the turbines to obtain

$$\sum_{i=1}^4 S_{bi} = \sum_{i=1}^4 S_{dhi} + \sum_{i=1}^4 S_{sti} \quad (24)$$

This is the total HP steam production that we want to produce at a minimum fuel energy consumption. The minimization problem could therefore be presented as

$$\begin{aligned} \min Q_{bk3} &= \sum_{i=1}^4 (a_{bi}n_{bi} + b_{bi}S_{bi}) \\ \text{where} \\ \sum_{i=1}^4 S_{bi} &= \sum_{i=1}^4 (a_{sti}n_{sti} + b_{sti}P_{sti} + c_{sti}P_{sti}^2 + d_{sti}n_{sti}S_{exti} + e_{sti}P_{sti}S_{exti}) \\ &\quad + \frac{1}{(1+\alpha)} \left\{ \sum_{i=1}^4 (a_{di}n_{di} + b_{di}W_{di}) - \sum_{i=1}^4 S_{exti}n_{sti} - \sum_{i=1}^4 S_{bpi}n_{di} \right\} \\ &\quad + \sum_{i=1}^4 S_{brti}n_{di} + \sum_{i=1}^4 S_{eji}n_{di} \end{aligned} \quad (25)$$

subject to the following constraints

$$\begin{aligned} P_{sti_{\min}} &\leq P_{sti} \leq P_{sti_{\max}} \\ W_{di_{\min}} &\leq W_{di} \leq W_{di_{\max}} \\ S_{exti_{\min}} &\leq S_{exti} \leq S_{exti_{\max}} \\ n_{sti} &= 0 \text{ or } 1 \\ n_{di} &= 0 \text{ or } 1 \end{aligned} \quad (26)$$

These constraints set the operating limits on the turbine and distiller loads and specify the operating status (as determined by the discrete variable n) of the different machines. This is again a mixed integer optimization problem in which some of the variables assume continuous values while the others are discrete variables which can only assume the values 0 or 1.

UAN West (Block 4)

Unlike Block 3, this block (see the schematic diagram in Fig. 5) does not consist of identical power generating or desalination units which can

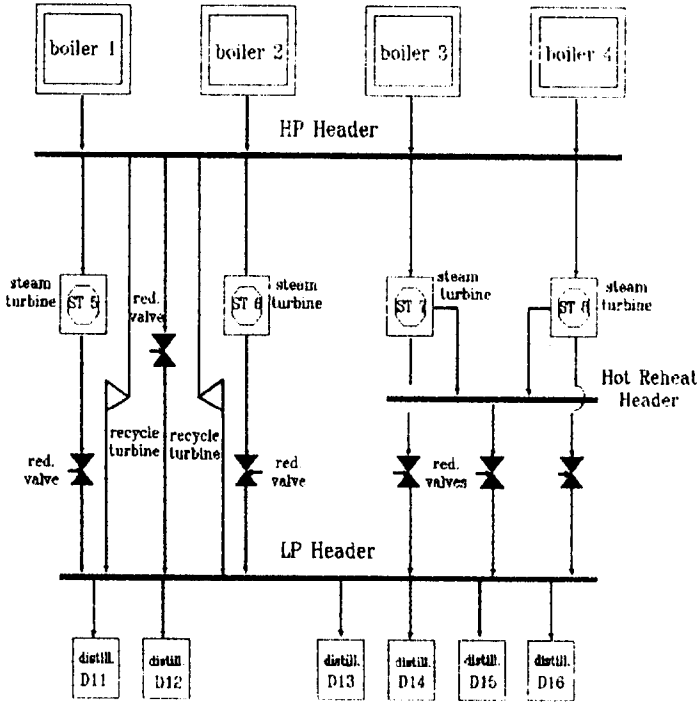


Figure 5. Simplified schematic of Block 4.

render the analysis somewhat more difficult. For example, the four steam turbines in Block 4 have different capacities and their design is also widely different. Two of the turbines, UANW 5 and 6, are controlled extraction/condensing non-reheat types and have a rated capacity of 60 MW. The other two turbines, UANW 7 and 8, are also of the controlled extraction/condensing type but are rated at 160 MW, with the additional design feature of steam reheat. Furthermore, the associated distillers are also not identical in design since distillers D11 and D12 have a rated capacity of 750 t/h (on polyphosphate additive) whereas distillers D13–D16 have a rated capacity of 950 t/h. We follow essentially the same procedure used before for Block 3 for obtaining the objective function and constraints. The fuel energy required by Block 4 to produce P_{bk4} MW of power and W_{bk4} t/h

of distillate is given by

$$\begin{aligned}
 Q_{bk4} &= \sum_{i=1}^4 Q_{bi} \\
 &= \sum_{i=1}^4 (a_{bi}n_{bi} + b_{bi}S_{bi})
 \end{aligned} \tag{27}$$

To evaluate the amount of steam required for the six distillers in Block 4, we should note first that two of the distillers, D11 and D12, have their brine recirculation pumps driven by small single-stage steam turbines, while the other four distillers, D13–D16, have their pumps driven by electric motors. The HP steam required by the recirculation pump turbines of distillers D11 and D12 should obviously be included in any steam mass balance.

The amount of LP steam required by the six distillers of Block 4 is supplied from three sources: turbine extraction steam, reducing station steam and steam discharged from the two brine recirculation turbines

$$\begin{aligned}
 \sum_{i=1}^6 S_{di} &= \sum_{i=1}^6 (a_{di}n_{di} + b_{di}W_{di}) \\
 &= \sum_{i=1}^6 S_{exti}n_{sti} + \sum_{i=1}^4 (1 + \alpha)S_{redi} + \sum_{i=1}^2 S_{brti}n_{di}
 \end{aligned} \tag{28}$$

The amount of HP steam to the reducing station, $\sum S_{red}$ can be obtained from this equation as

$$\sum_{i=1}^4 S_{redi} = \frac{1}{(1 + \alpha)} \left\{ \sum_{i=1}^6 (a_{di}n_{di} + b_{di}W_{di}) - \sum_{i=1}^4 S_{exti}n_{sti} - \sum_{i=1}^2 S_{brti}n_{di} \right\} \tag{29}$$

The total amount of HP steam required by the six distillers consists of steam to the reducing stations, steam to the brine recirculation turbines and steam to the distiller ejectors

$$\begin{aligned}
 \sum_{i=1}^6 S_{dhi} &= \frac{1}{(1 + \alpha)} \left\{ \sum_{i=1}^6 (a_{di}n_{di} + b_{di}W_{di}) - \sum_{i=1}^4 S_{exti}n_{sti} - \sum_{i=1}^2 S_{brti}n_{di} \right\} \\
 &\quad + \sum_{i=1}^2 S_{brti}n_{di} + \sum_{i=1}^6 S_{eji}n_{di}
 \end{aligned} \tag{30}$$

The amount of HP steam required by any of the turbines UANW 7 or 8 to generate a certain power depends on the mode of operation of these turbines. In addition to the pure condensing mode which is rarely used by these turbines since the attached distillers are most of the time in service, these turbines can operate in either pure extraction mode or combined

extraction/hot reheat mode. In the pure extraction mode, the amount of extracted steam (about 160 t/h) will be sufficient for two distillers. In the combined extraction/hot reheat mode, the amount of extracted steam will only be sufficient for one distiller, with the other distiller supplied by hot reheat steam via a reducing station.

With all four turbines operating in the extraction mode, the amount of HP steam required by the four turbines will be

$$\sum_{i=1}^4 S_{sti} = \sum_{i=1}^4 (a_{sti}n_{sti} + b_{sti}P_{sti} + c_{sti}P_{sti}^2 + d_{sti}S_{exti}n_{sti} + e_{sti}P_{sti}S_{exti}) \quad (31)$$

With the turbines UANW 5 and 6 running in the extraction mode while UANW 7 and 8 running in the combined extraction/hot reheat mode, the amount of HP steam required by the four turbines will be

$$\begin{aligned} \sum_{i=1}^4 S_{sti} &= \sum_{i=1}^2 (a_{sti}n_{sti} + b_{sti}P_{sti} + c_{sti}P_{sti}^2 + d_{sti}S_{exti} + e_{sti}P_{sti}S_{exti}) \\ &\quad + \sum_{i=1}^4 (a_{sti}n_{sti} + b_{sti}P_{sti} + c_{sti}P_{sti}^2 + d_{sti}S_{exti} + e_{sti}S_{hi}) \end{aligned} \quad (32)$$

The total amount of HP steam produced by the four main boilers of Block 4 can then be obtained by adding the HP steam for the distillers to that of the turbines

$$\begin{aligned} \sum_{i=1}^4 S_{bi} &= \sum_{i=1}^6 S_{dhi} + \sum_{i=1}^4 S_{sti} \\ &= \frac{1}{(1+\alpha)} \left\{ \sum_{i=1}^6 (a_{di}n_{di} + b_{di}W_{di}) \right. \\ &\quad \left. - \sum_{i=1}^4 S_{exti} - \sum_{i=1}^2 S_{bpi} \right\} + \sum_{i=1}^2 S_{bpi} + \sum_{i=1}^6 S_{eji} \\ &\quad + \left\{ \sum_{i=1}^2 (a_{sti}n_{sti} + b_{sti}P_{sti} + c_{sti}P_{sti}^2 + d_{sti}S_{exti} \right. \\ &\quad \left. + e_{sti}P_{sti}S_{exti}) + \sum_{i=3}^4 (a_{sti}n_{sti} + b_{sti}P_{sti} + c_{sti}P_{sti}^2 \right. \\ &\quad \left. + d_{sti}S_{exti} + e_{sti}S_{hi}) \right\} \end{aligned} \quad (33)$$

subject to the following constraints

$$\begin{aligned}
 P_{sti_{\min}} &\leq P_{sti} \leq P_{sti_{\max}} \\
 W_{di_{\min}} &\leq W_{di} \leq W_{di_{\max}} \\
 S_{exti_{\min}} &\leq S_{exti} \leq S_{exti_{\max}} \\
 S_{hi_{\min}} &\leq S_{hi} \leq S_{hi_{\max}} \\
 n_{sti} &= 0 \text{ or } 1 \\
 n_{di} &= 0 \text{ or } 1
 \end{aligned} \tag{34}$$

The constraints for Block 3 are also applicable for this block, but with the additional constraints on the limits of the amount of hot reheat steam used for desalination.

UAN West (Block 5)

Block 5 consists of two single-purpose power generating units, each one rated at 75 MW. The objective function and constraints for this block are

$$\begin{aligned}
 \min Q_{bk5} &= \sum_{i=1}^2 Q_{bi} \\
 &= \sum_{i=1}^2 (a_{bi}n_{bi} + b_{bi}S_{bi}) \\
 &= \sum_{i=1}^2 \{a_{bi}n_{bi} + b_{bi}(a_{sti}n_{sti} + b_{sti}P_{sti} + c_{sti}P_{sti}^2)\}
 \end{aligned} \tag{35}$$

subject to

$$\begin{aligned}
 P_{sti_{\min}} &\leq P_{sti} \leq P_{sti_{\max}} \\
 P_{bi_{\min}} &\leq S_{bi} \leq S_{bi_{\max}} \\
 P_{bk5} &= \sum_{i=1}^2 P_{sti} \\
 n_{sti} &= 0 \text{ or } 1 \\
 n_{bi} &= 0 \text{ or } 1
 \end{aligned} \tag{36}$$

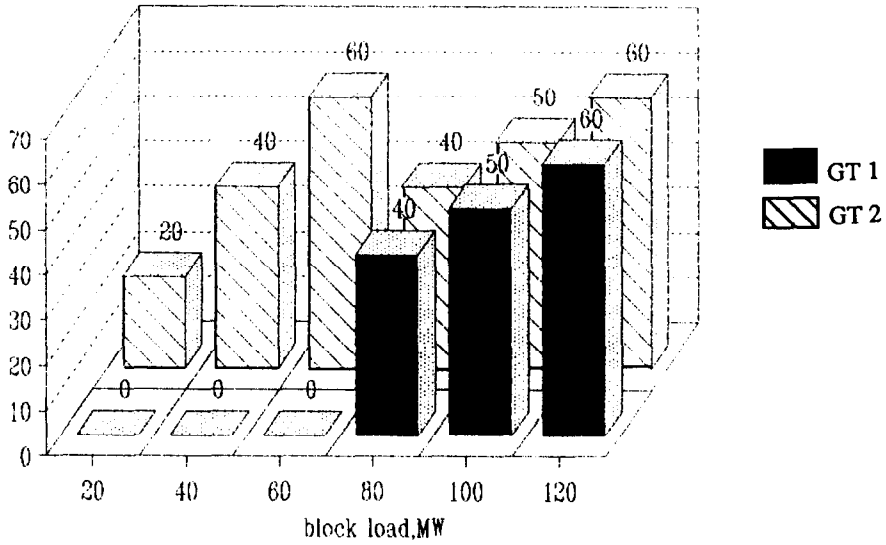


Figure 6. Unit load distribution for Block 1 with three distillers running.

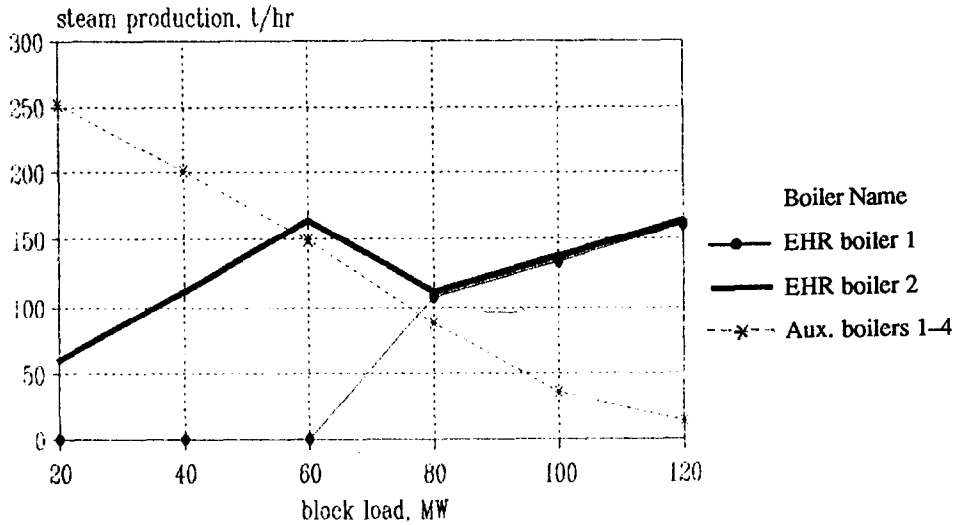


Figure 7. Steam production by EHR and auxiliary boilers with three distillers in operation.

1. Results

Block 1 (UANE)

The optimum power unit loading for Block 1 (UANE) with all the distillers running at full capacity is shown in Fig. 6. With the block power in the region below 60 MW (full-load of GT 1 and GT 2), the total load is carried by a single unit, with the other unit shutdown. Once the block power exceeds 60 MW, the power is split equally between the two units for a minimum block fuel consumption. The steam production by the EHR boilers and the auxiliary boilers for different Block 1 power generation is shown in Fig. 7 with all the distillers in full-load operation. For the case of no power generation by this block, all the steam required by the distillers (470 t/h) has to be supplied by the four auxiliary boilers since no steam is being produced by the EHR boilers. The amount of steam produced by any EHR boiler actually depends on the power generated by the gas turbine unit associated with that EHR boiler. Therefore as the Block 1 power generated increases, more steam will be produced by the EHR boilers and less steam will be produced by the auxiliary boilers. With the two gas turbines running at full capacity, the amount of MNP steam which can be produced by the two EHR boilers amounts to about 328 t/h, thus the shortfall which has to be supplied by the auxiliary boilers is estimated to be $470 - 328 = 142$ t/h which can be provided by a single auxiliary boiler. The optimum amount of steam to be produced by each auxiliary boiler of Block 1 with three distillers running at full load and for different levels of power generation is shown in Fig. 8. With no power generation, i.e. when the two gas turbines are shutdown, all the four auxiliary boilers have to be running at equal load, with each boiler supplying one fourth of the total amount of steam required by the three distillers. As the block power generation increases, less steam has to be produced by the auxiliary boilers since an increasing amount of steam is being produced by the EHR boilers. At a block power generation of 60 MW, three auxiliary boilers are in operation, and at 120 MW only two boilers are running. The total amount of steam produced by the auxiliary boilers of Block 1 will obviously be affected by the number of distillers in operation, as well as the block power generation. More distillers in operation results in more steam to be produced by the auxiliary boilers. This is reflected in Fig. 9, which shows the amount of steam produced by the auxiliary boilers at different block power levels when two and three distillers are running at full load. The optimal fuel energy consumption for Block 1 with three distillers running at full-load is shown in Fig. 10, in which the total fuel energy consumed by the gas turbines and the auxiliary boilers is shown plotted versus the Block 1 power generation. The optimum fuel energy consumption for this block was correlated with the block power generated for two and three distillers running at full

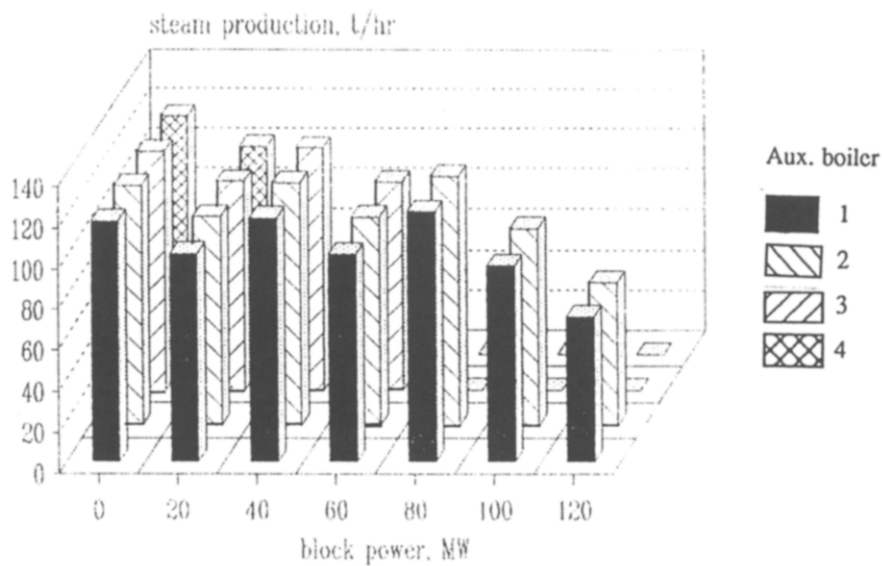


Figure 8. Steam production by auxiliary boilers with three distillers in operation.

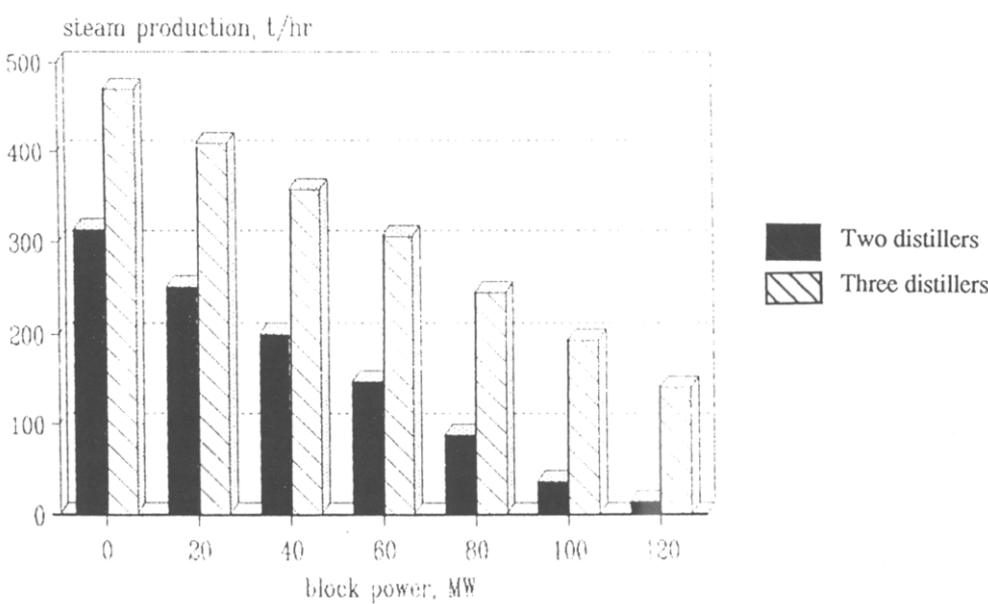


Figure 9. Steam production by auxiliary boilers with two and three distillers in operation.

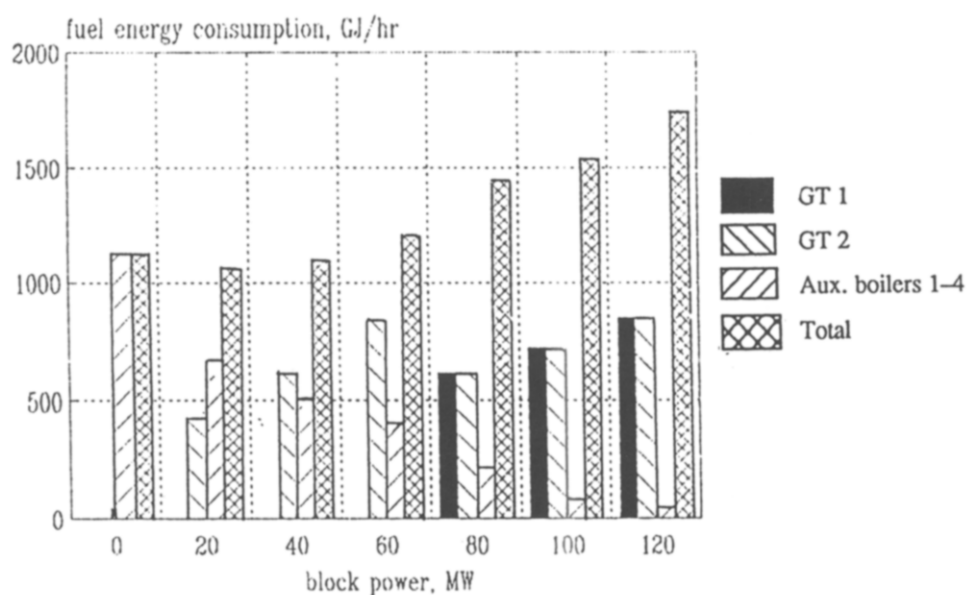


Figure 10. Optimum fuel energy consumption for Block 1 with three distillers in operation.

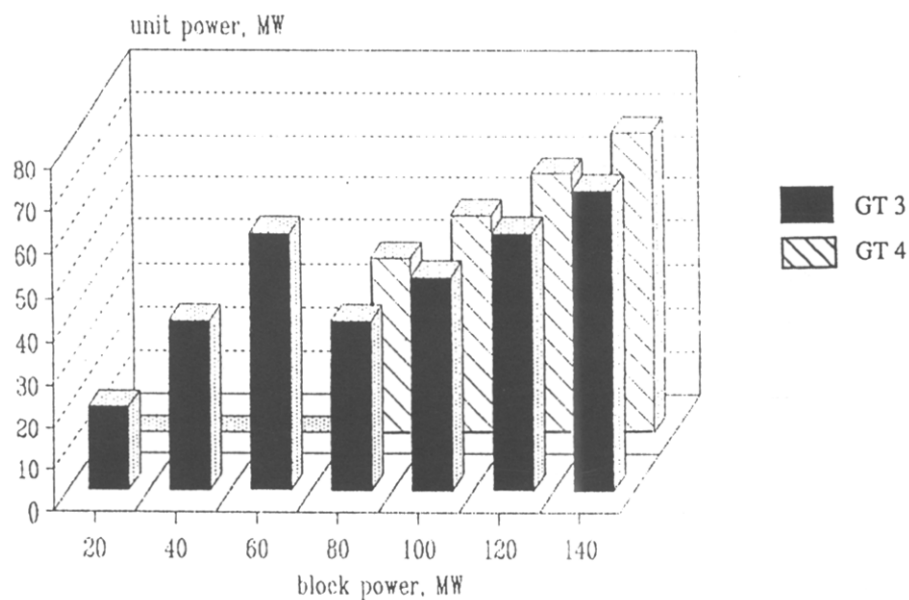


Figure 11. Unit load distribution for Block 2 and two distillers in operation.

load using the least-square method. With two distillers in operation the correlation takes the form

$$Q_{bk1} = 1118.6 - 6.73P_{bk1} + 0.187P_{bk1}^2 - 0.00074P_{bk1}^3 \quad (37)$$

With three distillers in operation the correlation is

$$Q_{bk1} = 1547.3 - 10.57P_{bk1} + 0.26P_{bk1}^2 - 0.0012P_{bk1}^3 \quad (38)$$

Block 2 (UANE ext.)

The power unit load distribution for Block 2 with two distillers in full-load operation is shown in Fig. 11 for the two gas turbines (GT 3 and GT 4) comprising this block. The optimum load distribution is similar to that of Block 1, namely, when the block power generated is equal to or less than the rated capacity of a single gas turbine, the load can most economically be carried by a single unit. However, if the block load exceeds the rated capacity of a single turbine, the load can then be split equally between the two turbines. With three distillers in operation, however, the block power will not be shared equally between the two gas turbines when the block power is greater than the rated power of a single turbine as can be seen from Fig. 12. The optimum steam production by the EHR boilers, resulting from

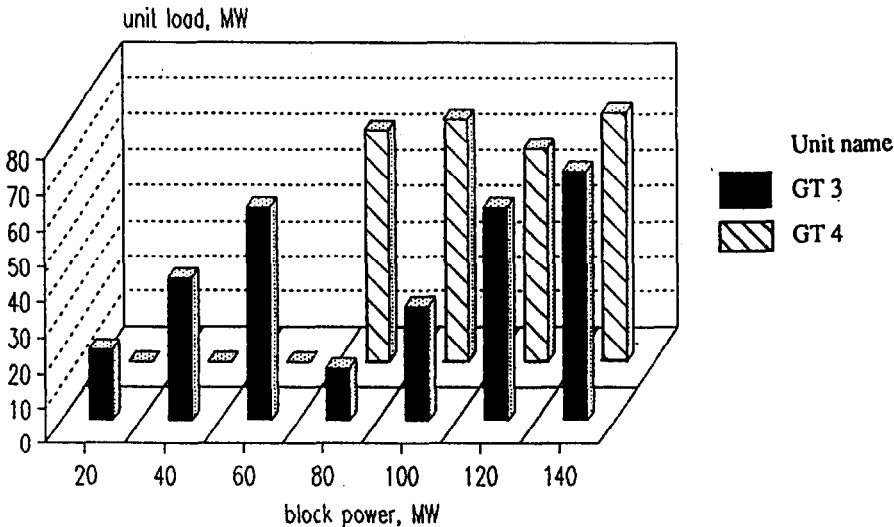


Figure 12. Unit load distribution for Block 2 with three distillers in operation.

both the exhaust gas streams as well as the supplementary firing, and the auxiliary boilers are shown in Fig. 13. It can be seen that as the block power

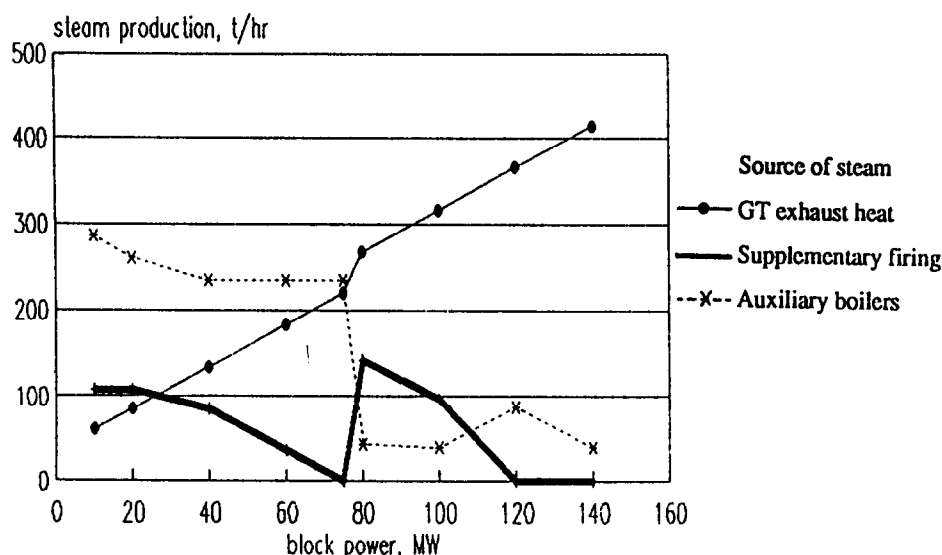


Figure 13. Steam production by EHR and auxiliary boilers with three distillers in operation.

increases from 10 MW to 75 MW (one gas turbine in operation), the steam generated by supplementary firing decreases from 107 t/h — corresponding to a power production of 10 MW — to zero at 75 MW. This variation is dictated by the limitation on the maximum steam production by the EHR boiler (210 t/h) and also by the limitation on the maximum supplementary fuel supplied which corresponds to a supplementary steam production of 107 t/h. As the block power increases beyond 75 MW (two gas turbines in operation), steam production by supplementary firing increases sharply, then drops gradually to zero as the block power continues to increase due to the same limitations mentioned above.

The optimum amount of steam produced by the auxiliary boilers reaches its maximum value at the minimum block power generated and gradually decreases as the block power increases. At a block power of 75 MW the amount of auxiliary boiler steam drops sharply as the second gas turbine is put into operation resulting in more EHR boiler steam becoming available. It is interesting to know the percentage contribution to the total steam production by each of the two gas turbine exhaust gas streams, the two supplementary firings and the two auxiliary boilers. This is clearly demonstrated in the four pie charts of Fig. 14 which gives the percentage contribution for a block power generation of 20 MW, 60 MW, 100 MW and 140 MW. One can notice that at low power levels (20 MW and 60 MW) most of the steam required by the three distillers is produced by the two auxiliary boilers. At high power level (100 MW and 140 MW) the exhaust

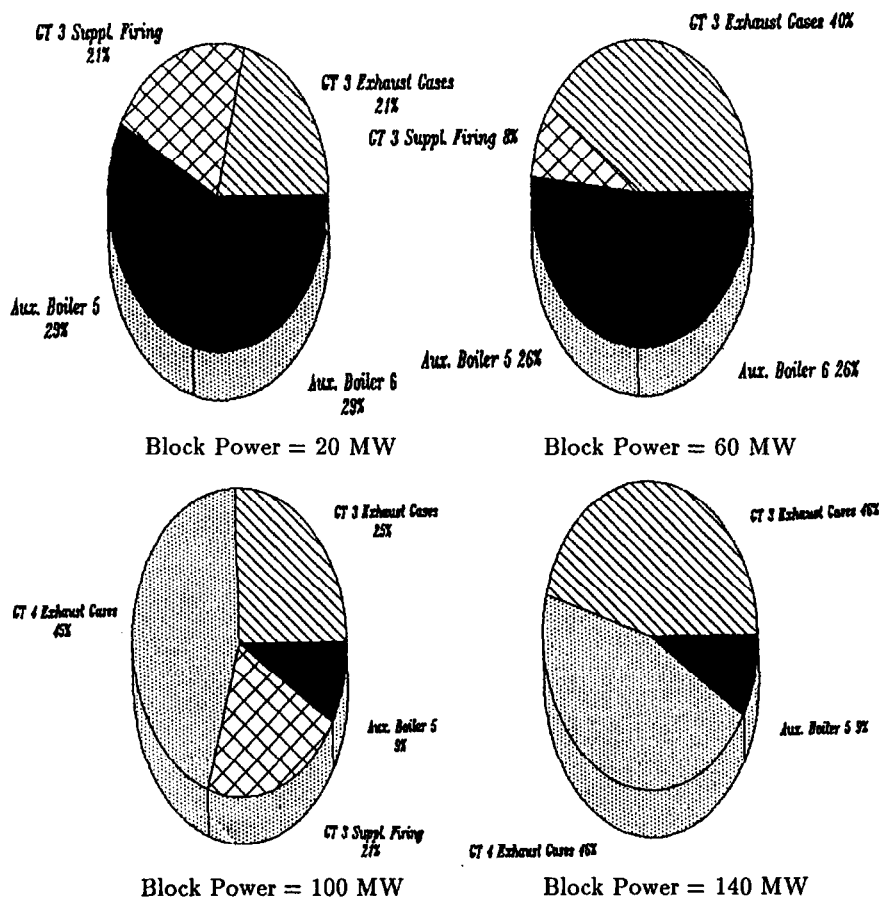


Figure 14. Percentage steam production by different sources with three distillers in operation.

gas streams are contributing most of the steam needed for desalination. The fuel energy required by Block 2 is shown in Fig. 15 with two and three distillers in full-load operation. The correlations which relate the fuel energy consumption in GJ/h to Block 2 power generation are as follows:

$$Q_{bk2} = 964.6 - 2.11P_{bk2} + 0.069P_{bk2}^2 - 0.00016P_{bk2}^3 \quad (39)$$

For three distillers operating at full load,

$$Q_{bk2} = 1339.0 + 0.323P_{bk2} + 0.034P_{bk2}^2 - 0.00013P_{bk2}^3 \quad (40)$$

Block 3 (UANW 1-4)

The optimum power unit load distribution for Block 3 with four distillers running at full load is shown (Fig. 16). With a Block power generation

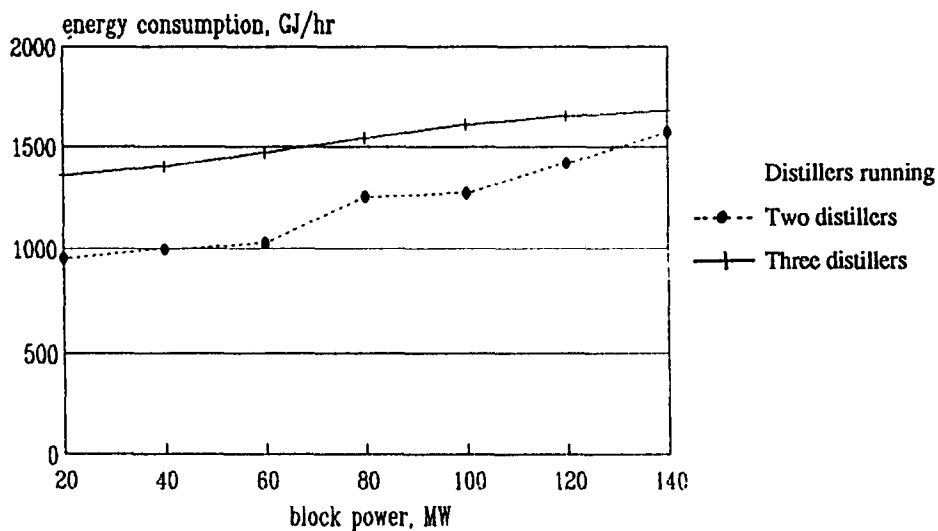


Figure 15. Optimum fuel energy consumption of Block 2 and two and three distillers in operation.

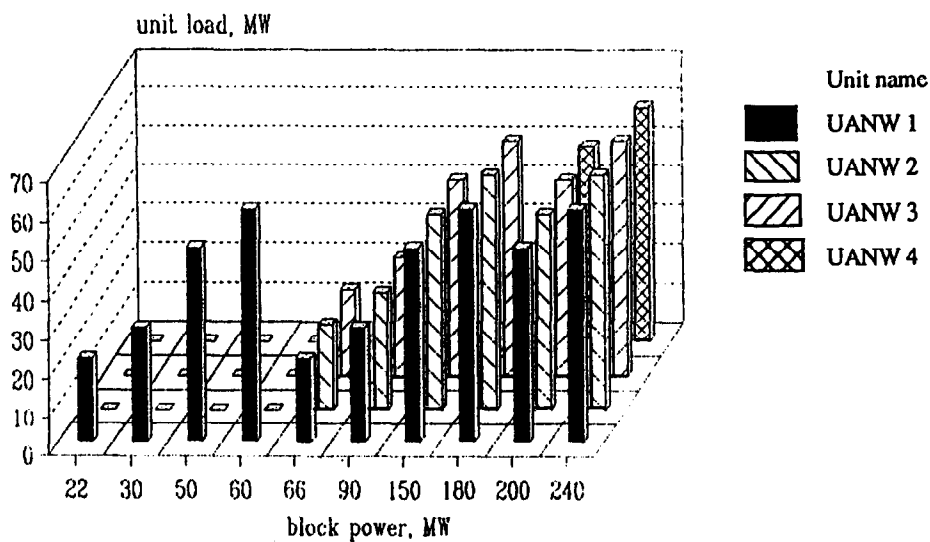


Figure 16. Unit load distribution for Block 3 with 4 distillers in operation.

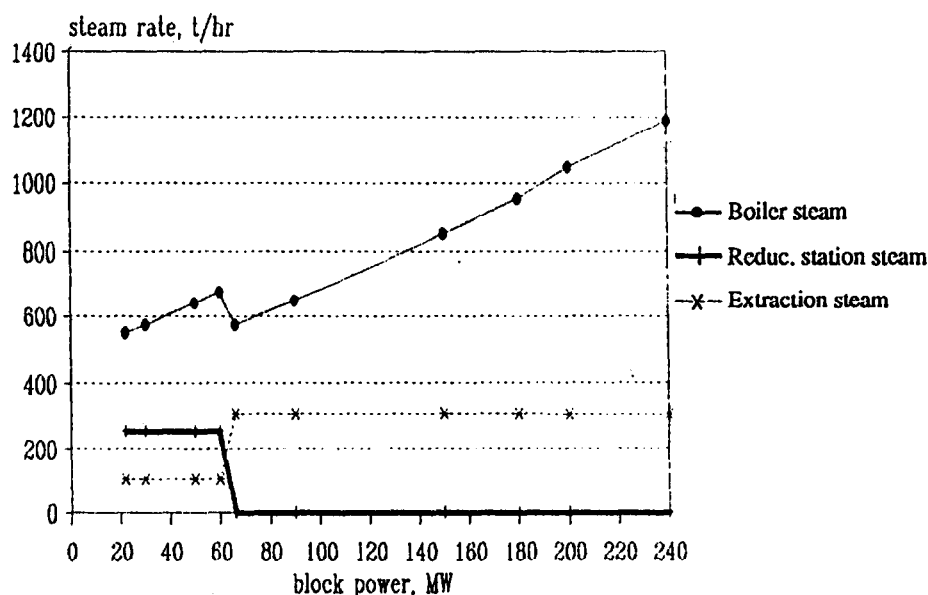


Figure 17. Steam production and consumption of Block 3 with four distillers in operation.

ranging between 22 MW (minimum load on a single turbine for stable operation) and 60 MW (maximum load), the block power is carried by a single turbine. In the block power range between 66 MW–160 MW the load is equally split between three turbines; with the block power above 180 MW, all the four turbines take an equal share of the load. Figure 17 shows the total steam production by the boilers, the steam flow to the reducing stations and the amount of turbine extraction steam versus the block power generation when all the distillers are running at full load. When one turbine is in operation (that is when the block load is below 60 MW), the extraction steam from this turbine is unable to satisfy the steam demand of the operating distillers, and the shortfall has to be supplied by reducing station steam. With a block power larger than 66 MW, three turbines will be in operation supplying enough extraction steam for all the distillers and there is no need for reducing station steam. The optimum fuel energy consumption by Block 3 with three and four distillers in full-load operation is shown in Fig. 18. The drop in the fuel consumption between 60 MW and 66 MW is due to the switching between a single turbine and three turbines with the accompanying drop in reducing steam requirement to zero. The correlations between the fuel energy consumption and block power generation was found to be as follows:

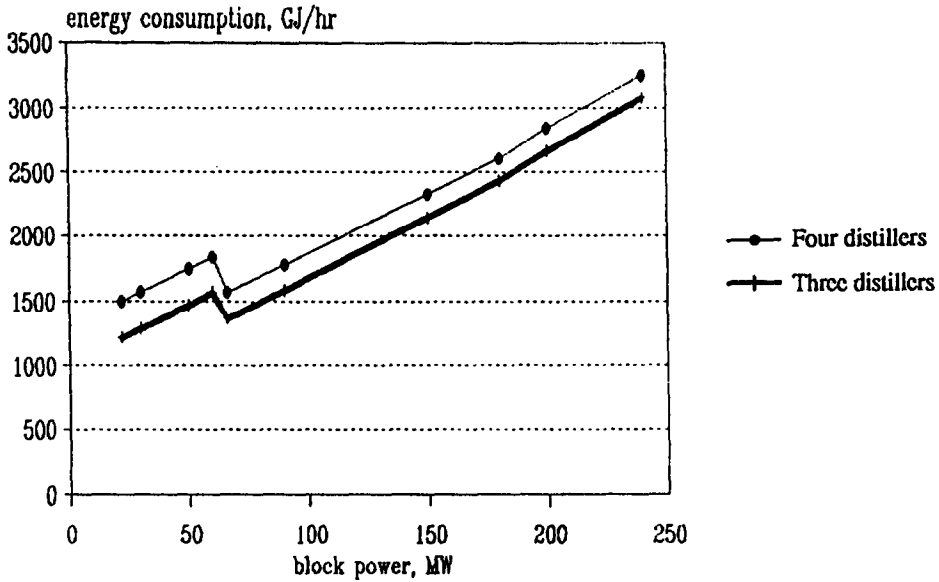


Figure 18. Optimum fuel energy consumption for Block 3 with three and four distillers.

Three distillers running at full load,

$$\begin{aligned}
 Q_{bk3} &= 1023.5 + 9.02P_{bk3} & 22 \leq P_{bk3} \leq 60 \\
 &= 738.3 + 9.6P_{bk3} & 66 \leq P_{bk3} \leq 240
 \end{aligned} \tag{41}$$

Four distillers running at full load,

$$\begin{aligned}
 Q_{bk3} &= 1297.3 + 9.0P_{bk3} & 22 \leq P_{bk3} \leq 60 \\
 &= 946.2 + 9.5P_{bk3} & 66 \leq P_{bk3} \leq 240
 \end{aligned} \tag{42}$$

Block 4 (UANW 5-8)

In the optimization of Block 4, the assumption was made that all the four steam turbines in this block will operate in extraction mode. This will exclude, for example, the hot reheat mode of operation for the turbines UANW 7 and 8. In the extraction mode, the minimum and maximum power that can be generated by this block are found to be about 200 MW and 460 MW, respectively. This limitation was included for two reasons, one is that the extraction mode is the normal operating mode for Block 4 since all the other modes involve additional fuel consumption, and the other is that the inclusion of all the operating modes in this work will substantially complicate the analysis.

The optimum power unit load distribution for this block is given in Fig. 19, which shows that at the minimum block load of 200 MW, each of the four steam turbine carries its minimum stable load which is 22 MW

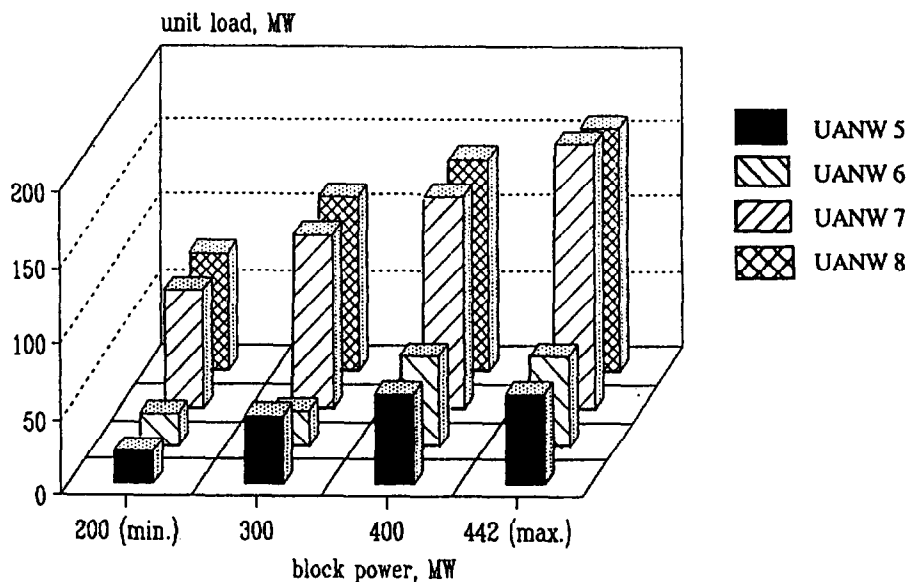


Figure 19. Unit load distribution for Block 4 with four distillers in operation.

for units UANW 5 and 6 and 78 MW for units UANW 7 and 8. As the block power increases, the load carried by all the units correspondingly increase with units 5 and 6 carrying loads that are close to each other, and similarly for units 7 and 8. The fuel energy consumed by this block when operating in the extraction mode with 4 and 6 distillers running at full load is shown in Fig. 20. The correlation between energy consumption and block power generated is as follows: Four distillers running at full load:

$$Q_{bk4} = 1963.2 + 3.60P_{bk4} + 0.0092P_{bk4}^2 \quad (43)$$

Six distillers running at full load:

$$Q_{bk4} = 2959.4 - 1.87P_{bk4} + 0.017P_{bk4}^2 \quad (44)$$

Block 5 (UANW 9 and 10)

As mentioned before, this block is a single-purpose power plant comprising two identical steam turbines each having a rated capacity of 75 MW. The optimization of this block is rather simple and results in the block power being carried by a single turbine when the block power is ranging

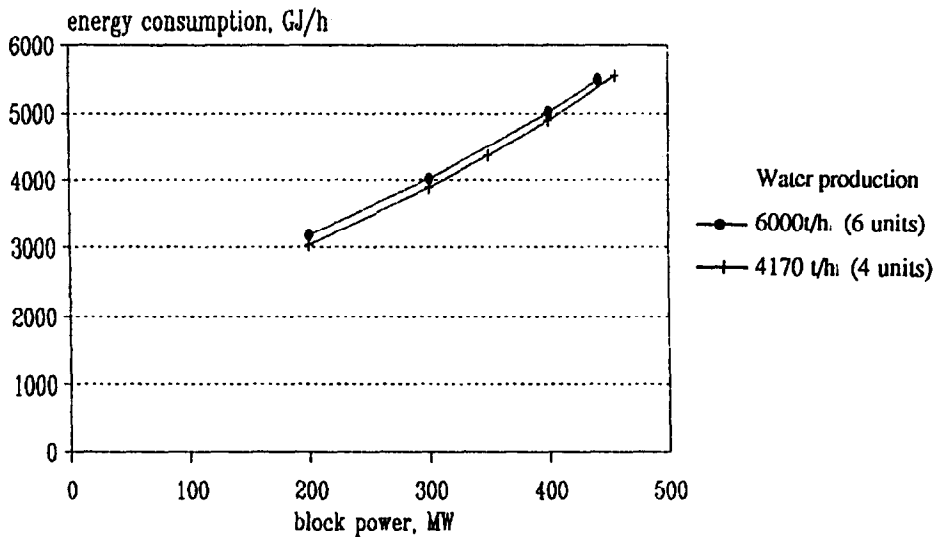


Figure 20. Fuel energy consumption for Block 4 with four and six distillers in operation with all the turbines in the extraction mode.

between 20 MW and 75 MW, and is shared equally between both units when the block power exceeds 75 MW. The correlation between the block fuel consumption and the block power generation was found to be

$$Q_{bk5} = 286.2 + 7.37P_{bk5} + 0.0169P_{bk5}^2 \quad (45)$$

Total block optimization

With the correlation functions for the five blocks now in hand, it is straightforward to optimize the five blocks together in order to find the optimum load distribution for the blocks that would minimize the total fuel consumption for the whole UAN station. The results are shown in Figs. 21 and 22 which give the optimum load carried by each block for different values of the total power generated by the UAN station when 16 and 11 distillers are in operation, respectively. At the minimum UAN station power generation of 300 MW, Block 1 will carry a load of about 35 MW. This increases gradually to the block full load (120 MW) when the station power reaches either 700 MW (when 11 distillers are in operation) or 600 MW (when 16 distillers are in operation), and is maintained at this power level thereafter. The pattern of change for Block 2 is essentially similar to that of Block 1 with the exception that the full load for this block (150 MW) is reached when the station power reaches either 500 MW (with 11 distillers in operation) or 400 MW (when 16 distillers are in operation).

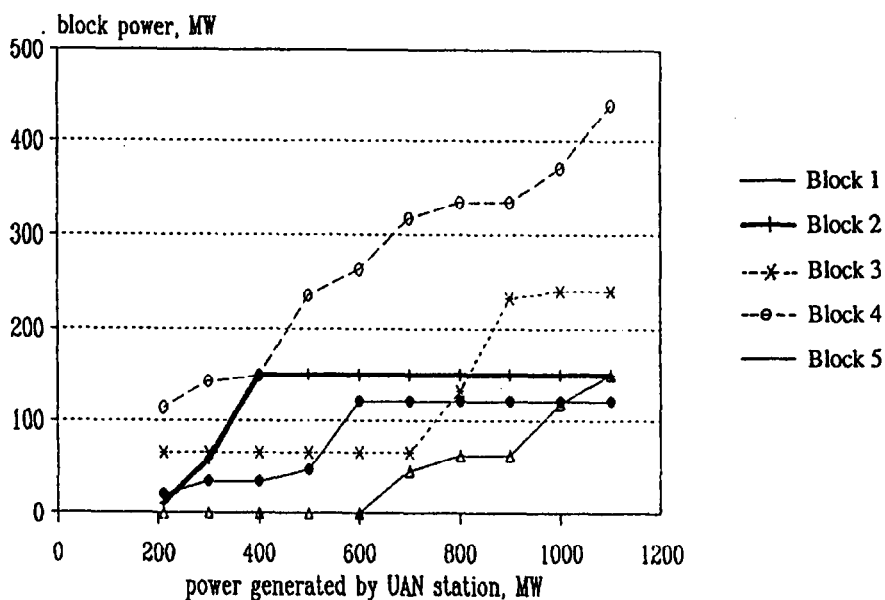


Figure 21. Optimum block load distribution with 16 distillers in operation.

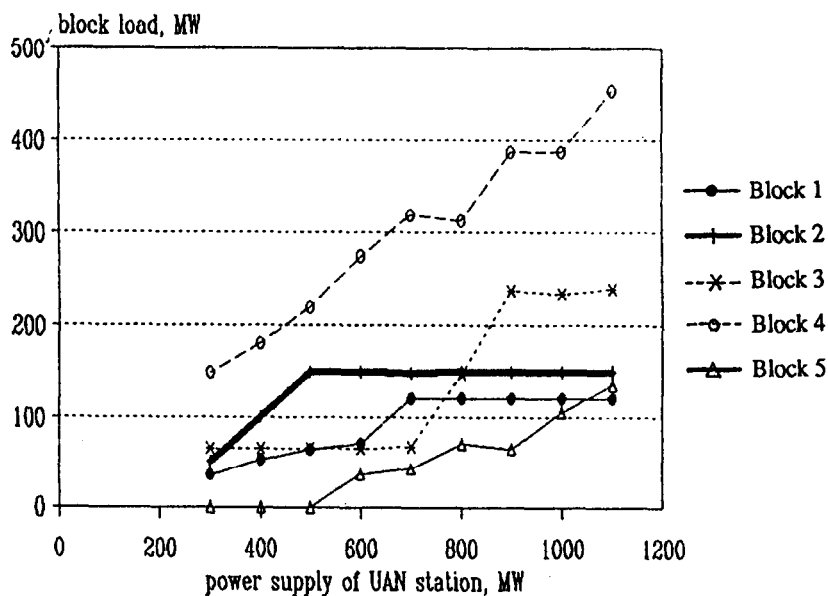


Figure 22. Optimum block load distribution with 11 distillers in operation.

Block 3 starts at 66 MW at the minimum station load, and continues with this load until the station power reaches 800 MW when the load on this block starts to increase until it reaches its rated capacity of 240 MW at a station power of about 900 MW. The block load then continues at its rated capacity until the maximum station power of 1100 MW. It may be noted that the minimum load for Block 3 is 66 MW which is generated by three steam turbines running at their minimum stable load of 22 MW and, at the same time, providing enough extraction steam for the distillers in operation. The maximum extraction steam that could be supplied by the three turbines, 3×103 t/h, would be sufficient to run four distillers at full load. It should be noted here that the mode of operation for Block 3 as well as Block 4 is the extraction mode which is the most efficient method of operation.

The load carried by Block 4 starts at about 150 MW at the minimum station power and gradually increases to nearly rated capacity (456 MW) at the maximum station power (1100 MW). It can be observed from Fig. 22 that Block 5 will remain shutdown as long as the UAN station power generation is less than about 500 MW (with 11 distillers) or 600 MW (with 16 distillers). This is because this block has no distillers associated with it. Since 11 or 16 distillers are assumed to be in service, priority of operation will naturally be given to the power generating blocks which have associated distillers in order to assure that these distillers will be in operation.

Th optimum total fuel energy consumption and heat rate of the UAN station is shown in Fig. 23 for the two cases where 11 and 16 distillers are

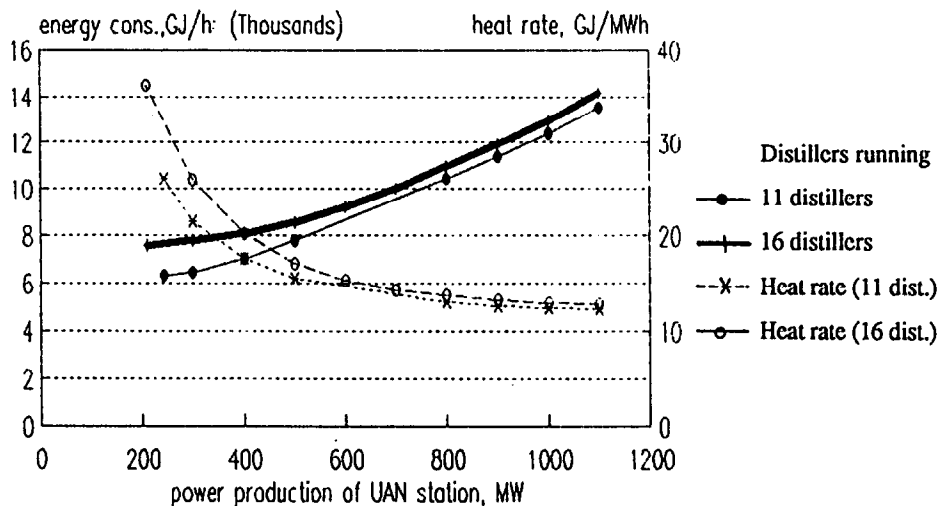


Figure 23. Optimum fuel energy consumption and heat rate for UAN station.

in operation. The heat rate is defined here as the fuel energy consumption in GJ/h per megawatt of electrical power produced. It can be seen that the heat rate experiences a gradual drop from its highest value at the minimum station power (300 MW) to its lowest value at the maximum station power (1100 MW). This is due to several additional inefficiencies resulting from operating the power generating units at part load compared to their full-load operation.

The heat rate values for the case where 11 distillers are in operation (Fig. 23) are slightly lower than when 16 distillers are in operation. This is due to the fact that for the first case (11 distillers in operation), the amount of extraction steam from the 8 controlled extraction/condensing turbines in Blocks 3 and 4 will be less than for the second case (16 distillers), which results in more steam expanding in the turbine and consequently more power being generated.

CONCLUSIONS

This paper shows how the economic dispatch of the UAN cogeneration station is carried out under purely static operating conditions in which electrical load variations are very slow and, at the same time, the water production by the MSF distillers is essentially constant. This idealistic situation is only an approximation to real systems but nevertheless can be looked at as a general guide on how the station load should be split between the different power generating units in order to minimize the station fuel consumption.

In carrying out this study, it was assumed that the normal operating mode for Blocks 3 and 4 is the extraction mode in which the major part of LP steam required for the distillers is supplied by bleed steam from the steam turbines. Under this mode of operation, the minimum and maximum power that could be generated from the UAN station is 300 MW and 1100 MW, respectively.

Based on the results of this study, the following conclusions may be drawn for the case where all the distillers are in full-load operation:

1. At the minimum station power, Block 1 should carry out about 11% of the power. This should increase to 20% (full load of Block 1) as the station power reaches 600 MW and then drops thereafter back to 11% at the maximum station power.
2. Block 2 carries about 19% of the total load at the minimum station power, and this should increase to 37% of the station power as the power increases to 400 MW (corresponding to full load of Block 2). With Block 2 running at full load from a station power of 400 MW till the maximum station power (1100 MW), the corresponding percentage load thus decreases from 37% to 14%.

3. Block 3 carries 22% of the minimum station load and its contribution should diminish to 11% at a station load of 700 MW. This corresponds to a 66 MW of power to be generated by Block 3 when the station power generation ranges between 300 MW and 700 MW. The load carried by this block increases to 26% at 800 MW and drops to 22% at the maximum station power (1100 MW).
4. The load carried by Block 4 at the minimum station power represents about 47% (corresponding to a block load of 114 MW) of this power and this drops down to about 40% (block load of 440 MW) at the maximum station power.
5. Block 5 starts to share a part of the station power only when this power reaches 700 MW. This block is therefore shutdown at any station power below 700 MW. At this station power the contribution of this block to the total load is only 5%, and this increases to 14% at the maximum station power.

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